Vishal John

M.D. Candidate, Class of 2028 · University of Louisville School of Medicine · Louisville, KY

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EDUCATION

Doctor of Medicine (M.D.) Candidate
University of Louisville School of Medicine
Jul 2024 – May 2028 (expected)
Louisville, KY

B.S., Biopsychology, Cognition, and Neuroscience
University of Michigan
Minor in Business Administration, Ross School of Business
Sep 2017 – May 2021
Ann Arbor, MI

RESEARCH EXPERIENCE

Clinical Research Technician
Koschmann Lab for Pediatric Brain Cancer, Michigan Medicine, Pediatric Hematology/Oncology
May 2021 – Jan 2023
Ann Arbor, MI

- Quantified circulating tumor DNA in CSF, plasma, and tissue from pediatric brain tumor patients by ddPCR to track response during ONC201 clinical trials.
- Developed a stem cell-derived thalamic and cortical organoid co-culture model of diffuse midline glioma invasion; presented as first author at the BioInnovation in Brain Cancer Symposium.
- Coordinated autopsy tissue and biofluid collection and built a barcoded tracking system for the lab's biorepository.
- Maintained IRB approval for human subjects studies and trained lab members in ddPCR, qPCR, western blot, organoid culture, cryosectioning, confocal microscopy, IHC, and ELISA.

Project Lead
miRcore (501(c)(3) nonprofit)
Jul 2023 – May 2024
Ann Arbor, MI

- Led the Citizen Scientist Sequencing Initiative, in which high school students sequence their own DNA and analyze variants of interest; presented this work at ASHG 2025.
- Directed a small RNA-seq research pipeline on Alzheimer's disease and mentored students in R, Linux, cutadapt, bowtie, samtools, and DESeq.
- Wrote and secured grant funding and managed partner and vendor relationships.

Research & Teaching Assistant
miRcore
Sep 2015 – May 2021
Ann Arbor, MI

- Taught high school students computational biology (GEO2R, STRING, R); mentored 68 students to publish research abstracts.
- Ran a 4-hour computational biology contest for 164 students from across the Midwest and California.

INNOVATION

Bluegrass Biodesign, Cardiology / Cardiothoracic Surgery Team
University of Louisville School of Medicine & J.B. Speed School of Engineering
2025 – 2026
Louisville, KY

- Identified an unmet clinical need through clinical observation and physician interviews, then designed, prototyped, and tested a device with an engineering team.
- Completed customer discovery and business training through UofL's NSF I-Corps program.

PUBLICATIONS

1. Wadden J, Adam T, Lau BC, …, **John V**, …, Koschmann C. Surgical waste from stereotactic glioma biopsies contains clinically relevant tumor DNA. *Scientific Reports (Nature)*. Oct 2026, Article in Press. doi:10.1038/s41598-026-75123-w
2. Venneti S, Kawakibi AR, Ji S, …, **John V**, …, Koschmann C (84 authors). Clinical efficacy of ONC201 in H3K27M-mutant diffuse midline gliomas is driven by disruption of integrated metabolic and epigenetic pathways. *Cancer Discovery*. Nov 2023, Vol 13(11), pp 2370-2393. doi:10.1158/2159-8290.CD-23-0131
3. Tripathy A†, **John V†**, Wadden J, Kong S, Sharba S, Koschmann C. Liquid biopsy in pediatric brain tumors. *Frontiers in Genetics*. Jan 2023, Vol 13, Article 1114762. doi:10.3389/fgene.2022.1114762
4. Wadden J, Ravi K, **John V**, Babila CM, Koschmann C. Cell-Free Tumor DNA (cf-tDNA) Liquid Biopsy: Current Methods and Use in Brain Tumor Immunotherapy. *Frontiers in Immunology*. Apr 2022, Vol 13, Article 882452. doi:10.3389/fimmu.2022.882452
5. Cantor E, Wierzbicki K, Tarapore RS, …, **John V**, …, Koschmann C. Serial H3K27M cell-free tumor DNA (cf-tDNA) tracking predicts ONC201 treatment response and progression in diffuse midline glioma. *Neuro-Oncology*. Aug 2022, Vol 24(8), pp 1366-1374. doi:10.1093/neuonc/noac030
6. Wadden J, Newell BS, Bugbee J, **John V**, …, Das R. Ultra-rapid somatic variant detection via real-time targeted amplicon sequencing. *Communications Biology (Nature)*. Jul 2022, Vol 5, Article 708. doi:10.1038/s42003-022-03657-6

PRESENTATIONS

1. **Vishal John**, Lucia Moscola, Ziqi Guo, Ava Dornseif, Cristina Castillo, Inhan Lee. From Base Pairs to Blood Type incorporating Machine Learning Program: High School Students Decode Their Own ABO Gene towards Privately-held Sequencing and AI in Biology. Poster, *American Society of Human Genetics (ASHG) 2026 Annual Meeting*, Montréal, QC, Canada. October 20–24, 2026.
2. **Vishal John**, Veena Thamilsevan, Lucia Moscola, Inhan Lee. Engaging citizen scientists through WES self-data analysis training. Abstract, *American Society of Human Genetics (ASHG) Annual Meeting*, Boston, MA. October 16, 2025.
3. **Vishal John**, Lucia Moscola, Ava Dornseif, Ziqi Guo, Inhan Lee. From Blood Types to Base Pairs: High School Students Decode Their Own ABO Gene. Abstract, *Research!Louisville*, Louisville, KY. October 14, 2025.
4. Rebecca W. Gao, Jack Wadden, Robert Dickson, Rayan Aittaleb, **Vishal John**, Carl Koschmann, Steven Chinn. Nanopore Sequencing for the Real-time Detection of Melanoma-Involved Lymph Nodes. Oral presentation, *AAO-HNSF 2023 Annual Meeting & OTO Experience*, Nashville, TN. September 30 – October 4, 2023. doi:10.1002/ohn.435
5. **Vishal John**, Rodrigo Cartaxo, Robert Doherty, …, Carl Koschmann. Use Of A Thalamic Organoid Co-culture Model To Determine Drivers Of Diffuse Midline Glioma Migration And Invasion. Poster, *Inaugural Brain Cancer Symposium*, Ann Arbor, MI. September 30 – October 1, 2022.
6. Jack Wadden, **Vishal John**, Kait Verbal, Amy Bruzek, Wajd Al-Holou, Jason Heth, Hugh Garton, Carl Koschmann. Zero-waste Molecular Diagnostics From Biopsy Needle Wash Water. Poster, *Pediatric Research Symposium*, Ann Arbor, MI. May 24, 2022.
7. Amy Bruzek, Jack Wadden, **Vishal John**, Kait Verbal, Wajd Al-Holou, Jason Heth, Hugh Garton, Carl Koschmann. Ultra-rapid Molecular Diagnosis of H3K27M Mutation from Biopsy Washings: A Feasibility Report. Poster, *AANS/CNS Pediatric Neurosurgery Section Meeting*, Washington, DC. December 1–4, 2022.

OTHER PUBLICATIONS

John V, Patel N. *The Brave Little Immune Team*. Children's book introducing the immune system to ages 4–8. 2025. Available on Amazon.

LEADERSHIP

President — Jul 2025 – Present
Oncology Interest Group, University of Louisville School of Medicine — Louisville, KY

President — 2025 – 2026
South Asian Medical Student Association, University of Louisville School of Medicine — Louisville, KY

- Planned and budgeted a year of seven cultural, professional, and wellness events, including Diwali and Holi celebrations and a Ramadan Iftar.
- Organized a South Asian physician mentorship meet-and-greet, with specialties chosen by member poll, and a physician seminar co-hosted with a specialty interest group.
- Launched a yoga and chai event introducing first- and second-year students to the practice's Indian roots as a way to decompress from class.

Peer Mentor — 2025 – 2026
University of Louisville School of Medicine — Louisville, KY

- Mentored three first-year medical students through the transition to medical school, advising on study strategies, exam preparation, wellness, and getting involved in research and student organizations.

Leadership Team — Feb 2022 – Present
Rock Cancer — Ann Arbor, MI & Louisville, KY

- Free rock climbing for children and young adults with cancer. Built the program's website and merchandise shop and helped grow it to national coverage on NBC Nightly News and mlive.
- Helped launch the Louisville program in partnership with Norton Children's Cancer Institute; volunteer belayer.

President — 2018 – 2021
Genes in Diseases and Symptoms (GIDAS), University of Michigan — Ann Arbor, MI

- Raised over $27,000 for disease research microgrants and organized a monthly speaker series with physicians and scientists.

CLINICAL & COMMUNITY SERVICE

Volunteer — 2025
APAMSA Louisville Community Health Fair — Louisville, KY

- Supported a free screening fair (blood pressure, diabetes, vision) for Louisville's Asian Pacific community, entertaining children with balloon art so families could be screened.

Volunteer — Feb 2022 – Present
Hospital Elder Life Program (HELP), Michigan Medicine — Ann Arbor, MI

- Supported hospitalized older adults with orientation, mobility, feeding, and sleep protocols to prevent delirium.

Food Server — Dec 2021 – Present
The Breakfast Program at Saint Andrew's — Ann Arbor, MI

- Serve hot breakfasts, bagged lunches, and clothing to Ann Arbor's unhoused and food-insecure community.

HONORS & AWARDS

James B. Angell Scholar, University of Michigan — 2020

South Asian Studies Fellow, U-M Center for South Asian Studies — 2019

Center for Japanese Studies Scholarship, University of Michigan

University High Honors (5x), University of Michigan

Health Sciences Scholars Program, University of Michigan 2017 – 2018

CERTIFICATIONS

Stop the Bleed, American College of Surgeons 2025

SKILLS & INTERESTS

Laboratory	ddPCR, qPCR, western blot, 2D and organoid cell culture, cryosectioning, confocal microscopy, immunohistochemistry, ELISA, primer/assay design
Computational	R, Linux/bash, RNA-seq analysis (cutadapt, bowtie, samtools, DESeq), high-performance computing
Languages	English (native), Malayalam (fluent)
Interests	Rock climbing, balloon art, coffee, children's science writing